Линия второго порядка называется множество

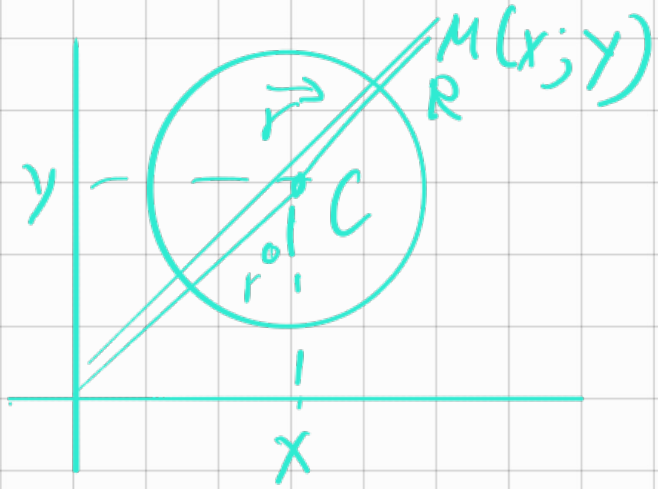
$$a_{11}x^2 + a_{12}y + a_{22}y^2 + 2b_1x + 2b_2y + c = 0$$

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

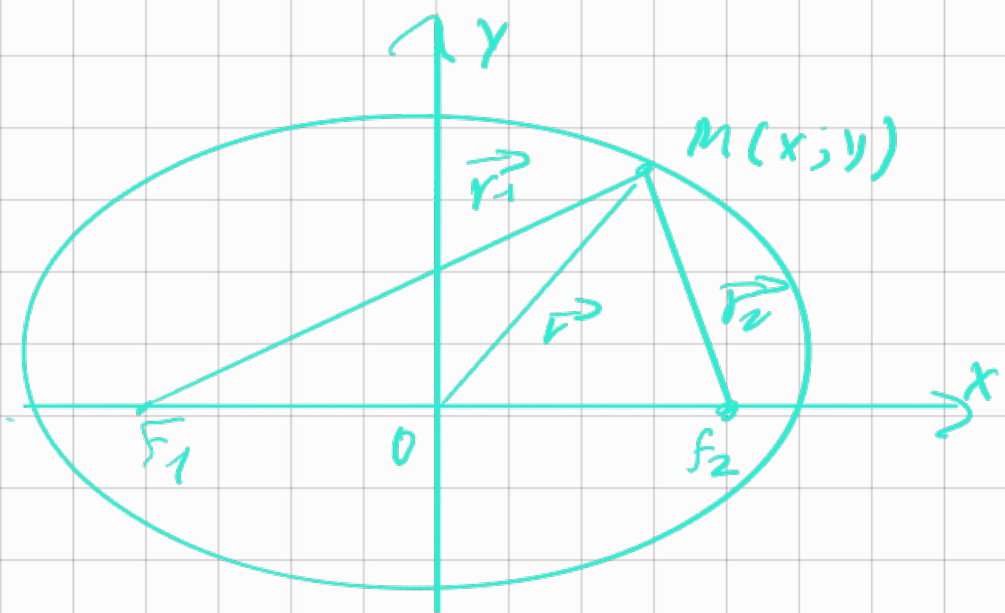
Если

$A = B = C = 0$, то $Dx + Ey + F = 0$

Окружность



$$|\vec{r} - r_0| = R$$



$\{M \mid |F_1\vec{M}| + |F_2\vec{M}| = 2a\}$

$|\vec{r} - \vec{r_1}| + |\vec{r} - \vec{r_2}| = 2a$

$|\vec{F_1M}| + |\vec{F_2M}| = const$

$|\vec{F_1M}| + |\vec{F_2M}| = 2a$

$|\vec{F_1F_2}| = 2c$

$\vec{F_1M} = \{x + c, y\}$; $\vec{F_2M}$ $\{x - c\}$ y^2

ПРАКТИКА

$x^2 + y^2 - 4x + 8y - 16 = 0$

$x^2 - 4x + 4 - 4 + y^2 + 8y + 16 - 16 - 16 = 0$

$(x+2)^2 + (y+4)^2 = 6^2$

Ответ: $(2; -4)$ $R = 6$

$9x^2 + 9y^2 + 42x - 54y - 95 = 0 \quad / \ 9$

$x^2 + \frac{14}{3}x + \left(\frac{7}{3}\right)^2 + y^2 - 6y + 9 - \frac{49}{9} - 9$

$$-\frac{95}{9}=0$$

$$\left(x+\frac{4}{3}\right)^2+(y-3)^2=25$$

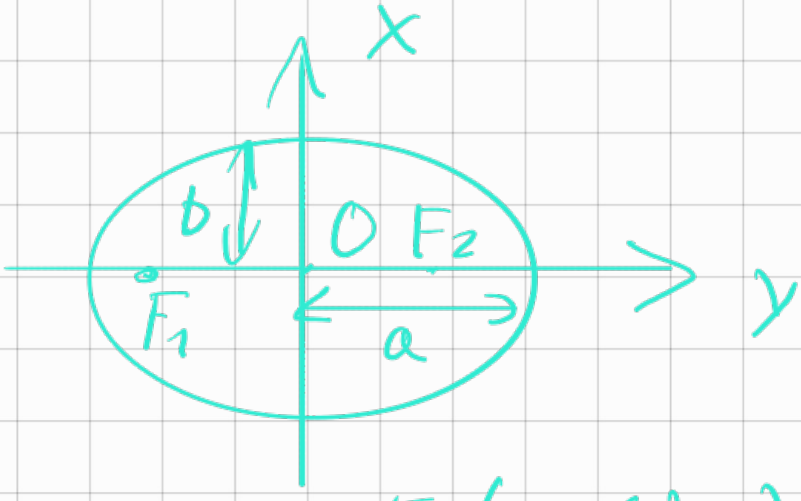
$(-1;3)$ $(0;2)$ $(1;-1)$

$$(x-a)^2+(y-b)^2=R^2$$

$$(-1-a)^2+(3-b)^2=R^2$$

$$\begin{cases}(-1-a)^2+(3-b)^2=R^2\\ a^2+(2-b)^2=R^2\\ (1-a)^2+(-1-b)^2=R\end{cases}$$

Эллипс



a – большая полуось

b – малая полуось

$F_1(-c;0)$ $F_2(c;0)$

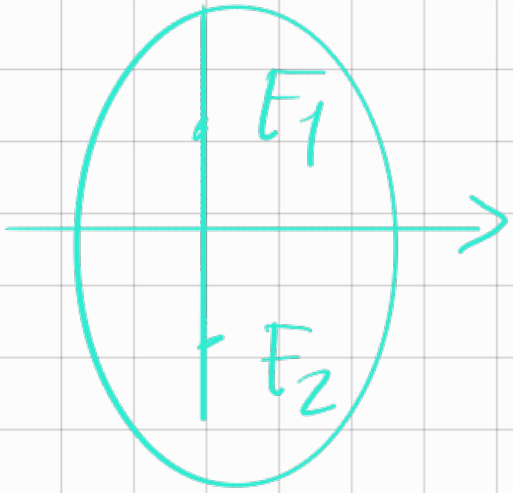
$$c^2 = a^2 - b^2$$

$r_1 = a + \varepsilon x$

$r_2 = a - \varepsilon x$

$\varepsilon = \frac{c}{a}$ эксцентриситет

$x = \frac{a}{\varepsilon}$ $\quad x = -\frac{a}{\varepsilon}$



$b > a$

$c^2 = b^2 - a^2$

$\varepsilon = \frac{c}{b}$

$4x^2 + 3y^2 - 8x + 12y - 32 = 0$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$4(x^2 - 2x) + 3(y^2 + 4y) - 32 = 0$

$4(x^2-2x+1-1)+3(y^2+4y+4-$

$4)-32 = 4(x-1)^2-4+3(y+2)^2$

$-12 \quad -32$

$4(x-1)^2+3(y+2)^2=48$

$\frac{(x-1)^2}{12}+\frac{(y+2)^2}{16}=1$

$24x^2+48y^2=1176$

$2(12x^2)+4(4y^2)=1176$